Architecture of Complex Systems

Week 2: Function and Emergence

Project Portfolio

Name

Martin Santiago Benavides Ardila

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Instructions

Before you begin, you should save your Project Portfolio on your local drive. We recommend the following format:

Lastname_Firstname_Course1_Week2

Please note: You will not be able to re-download your file after submission; therefore, please keep this file in a central location for future reference.

The project deliverable is an **individual submission**. A scoring rubric can be downloaded from the “Project Instructions” page for this week.

You will be self-assessing your work. If you have any questions, feel free to start a thread in the Discussion Forum.

Note: edX has a 10MB file size limit for document submission. If you have selected large image(s), you may need to resize before submitting, OR you may simply include a web URL for the image in the image location. Be sure to submit your assignment at least one hour before the deadline to provide time for troubleshooting.

Once the deadline passes, you will not be able to upload the document and therefore will not be able to submit and complete the assignment.

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Week 2 Project

Overview

In the second project activity of this course, you will build on your project work from Week 1 by developing a graphical representation of the system.

The steps to the right will guide you through this process.

REQUIRED STEPS:

Step 1: List object elements or abstractions of form.

Step 2: Identify system operands and functions.

Step 3: Develop an OPM diagram for the system.

Step 4: Describe how functional information is normally conveyed in your field. Detail an example.

Step 5: Review and submit your project.

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STEP 1: ABSTRACTIONS OF FORM

For the system you chose in Week 1, list five or more object elements or abstractions of form to make a Level 1 decompositional view. Don't feel constrained to use the same objects as you listed in Week 1.

Object elements or abstractions of form:

Prime number

- Input (n) ` - print loop
- Array primes[ ]
- Initialization loop
- Nested loop conditions.

Describe how and or why you used these elements / abstractions of form to construct your graphical decompositional view for the form of your system.

These elements were selected because they represent the Level 1 of system decomposition, as shown in the course material from Week 1. They are the core structural components that capture both the form and function of the algorithm. Their relationships and interactions reveal how external function of the system emerges once the Level 1 elements are combined.

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STEP 2: SYSTEM OPERANDS AND FUNCTIONS

For your next step, you will consider value related and principal internal operands and states. With your chosen system in mind, answer the following questions:

What is the value related operand? What are the value related states that change? What is the value related process of changing those states?

Operand: integer number States that change: Classification of each number (prime or not)
Process: search

What are the principal internal operands? What principal internal processes act on them? What are the principal internal functions?

Prime numbers array / initialization, calculating, looping, comparison, appending, printing. /
assigning an initial prime status, filtering numbers, testing numbers, appending the number, printing prime array.

How do the principal internal functions connect to form the primary value pathway? How does the external function emerge from these internal functions?

Internal functions: connected sequentially through the primes array by flagging-> filtering -> selecting.
External functions: emerges from finding and printing prime numbers.

How do the operands change because of forms?

The cells of the primes array change from 1 to 0 depending on the range and condition of each loop. The initialization loop sets all to 1, the sieve loop clears multiples to 0, and the print loop reads without modifying.

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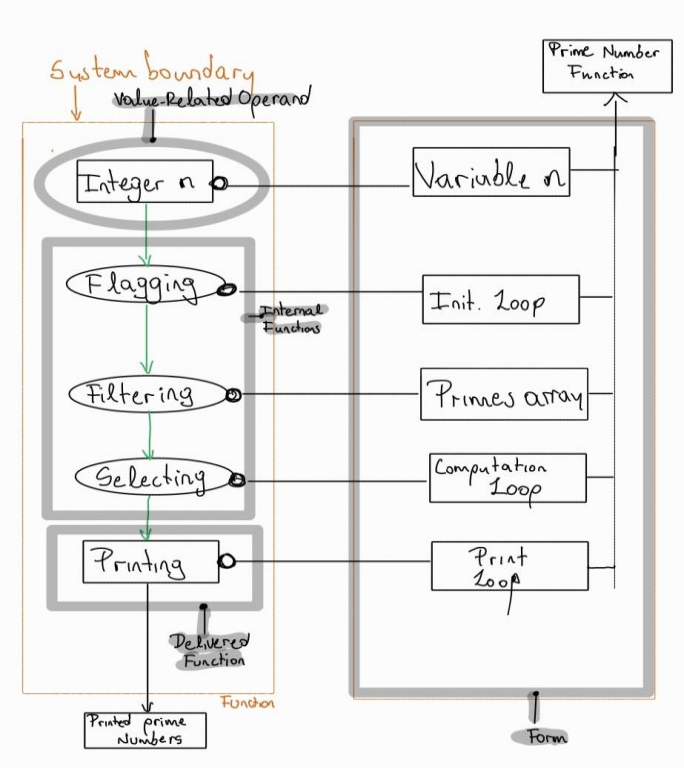
STEP 3: DEVELOP AN OPM DIAGRAM

For your system, develop an OPM diagram and insert the diagram below. Highlight or circle and label the following: value related operand, delivered function, internal functions (operands and processes), and form. Provide a brief description of each in the field provided in the next slide.

The following syntax should be used within the OPM diagram: System Boundary, all lines labeled, only OPM symbols used, no object-object links unless clearly explained why, no process-process links unless clearly explained why.

Please remember the file size limit and resize or paste the image URL instead, as needed.

System Diagram/Schematic



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STEP 3: DEVELOP AN OPM DIAGRAM

Provide a brief description of each in the field provided of the following: value related operand, delivered function, internal functions (operands and processes), and form.

Value Related Operand:

Integer n: input number that determines the size of primes array

Delivered Function:

Finding and printing all prime numbers up to n. Emerges from internal funcionts.

Internal Functions (operands and processes):

Operands: the primes array and the index variables i and j.

Processes: flagging (sets all cells to 1), filtering (overwrites multiples to 0 using the sieve), and selecting (reads and outputs cells that remain 1).

Form:

Variable n, primes[ ] array, initialization loop. sieve computation loop, and print loop. Each structural element enables its corresponding internal process and position of each loop directly determines how the primes array is written, updated, and read.

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STEP 4: FUNCTIONAL INFORMATION

For your last step, you will think about how functional information is normally conveyed in your field or discipline. Are processes indicated? Operands? Are processes and operands combined into functions? In your description, be sure to cite and briefly describe at least one specific example.

In physics, and particularly in quantum optics, functional information is conveyed through schematic diagrams that represent both the experimental setup and the logical flow of the system. These diagrams are essential because they communicate simultaneously the form of the optical components such as beam splitters, mirrors, detectors, and waveplates and the function of the scheme, meaning how photons or quantum states are manipulated and measured across the system.Processes are explicitly indicated through the path of the optical field and the sequence of interactions with each component. Operands appear as the quantum states or photon modes that are transformed at each stage. Processes and operands are combined into functions when the full scheme is read as a whole.

For example: in my current project about orbital angular momentum (OAM) manipulation using optical elements, the schematic diagram shows how a photon carrying a specific OAM mode enters a sequence of optical components such as spiral phase plates or spatial light modulators, undergoes a controlled mode transformation at each stage, and exits in a target OAM state. The delivered function reproducing an X-gate operation for quantum computing only becomes readable when the full optical path is traced as a complete sequence. No individual component performs the gate such as the function emerges from the combined and ordered interaction of all elements with the photon's OAM degree of freedom.

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STEP 5: REVIEW & SUBMIT PROJECT

- Submit and self assess your completed Week 2 Project Portfolio file
- Note: The maximum file size that can be submitted is 10MB.
  - A sample project submission and scoring rubric can be downloaded from the “Project Instructions” page for this week.
  - Please remember that there are two parts to this assignment: Submission and self assessment. Please be sure to provide enough time to complete both parts.

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